Import Price Shocks, FX-Monetary Policies, and Real Income Stabilization

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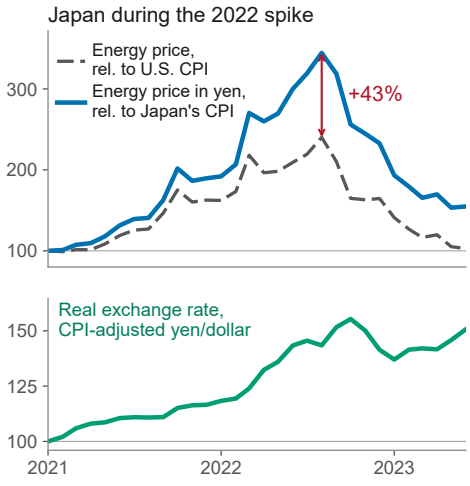
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Essential Import Price Spikes

- Recent years: **import-dependent economies** facing recurrent episodes of large, temporary spikes in essential-import prices:
  - 2022: LNG and crude +100-500%
  - 2026 Q1: crude +60%; Asian LNG +140%
- For an import-price-taking economy, this is a **negative terms-of-trade** shock: give away more domestic output to buy the import. Price-inelastic demand: shock translates into **real-income losses**.
- In the data: concurrent **real depreciation** endogenously worsens the terms of trade and amplifies real-income losses.
- Policy concerns in a monetary economy: **imported inflation** and nominal depreciation.

How should monetary and exchange rate policies respond?



Existing Policy Views

"It is far simpler to allow one price to change, namely, the price of foreign exchange. . . "

Friedman (1953, p. 173)

- **Representative-agent small-open-economy NK framework**:
  - **FX policy: laissez-faire.** Leave capital flows undistorted; **let the real exchange rate depreciate** freely, as a shock absorber.

    Rationale: with perfect access to international asset markets, a transitory import price spike is only a PDV loss. Households borrow and smooth it over time.
  - **Monetary policy: targets nominal rigidities.** Under rigid wages: stabilize wage inflation and activity; let prices rise so real wages fall.
- **TANK/HANK**: **monetary policy protects real incomes** of borrowing-constrained households through activity: **less contractionary than RA**. FX response stays laissez-faire.

The Policy Set

Monetary policy

- Operates through nominal rigidities
- Influences activity: output, employment
- Money growth, nominal rate

FX policy

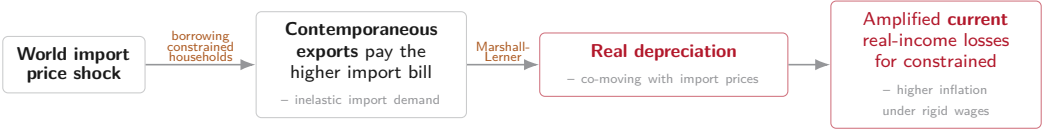
- Distorts international financial flows
- Influences real exchange rate path
- Capital control taxes, FX interventions

- Typically: FX policy for managing volatile capital flows, financial crises, macroprudential policy, and ToT manipulation.
- Monetary policy for fighting inflation and **protecting real incomes** of constrained households.

This Paper

- Open economy framework.
  - Import-dependent economy, small in import and asset markets. Large in export markets.
  - **Marshall-Lerner condition**: Inelastic import demand. Downward-sloping, price-elastic export demand: **real depreciation** required for higher export revenues.
  - **Heterogeneous households**: some borrowing constrained: cannot smooth consumption through the import price shock.
  - **No direct fiscal transfers** to constrained households.

From shock absorber to shock amplifier:



How does heterogeneity change the roles of monetary and FX policies? To what extent can they protect households who cannot borrow through the shock?

Main Result: FX Policy and Real Income Stabilization

- A laissez-faire "let it depreciate" exchange rate response is costly for constrained households.
- Instead, policy *should* **lean against a contemporaneous depreciation** to **protect their real incomes** during shock episodes.
- **Active role for FX policy** under household heterogeneity in managing the real exchange rate path during episodes of temporary spikes in essential-import prices and associated inflation.

FX policy and indirect real income effects:



Main Result: FX Policy and Real Income Stabilization

- **The real-income stabilization channel of exchange rate policy**:
  - Induces stronger capital inflows during a spike and makes foreign purchasing power ("dollars") more abundant in the FX market.
  - Finances more of the current import bill through capital inflows and less through current exports: retains more output at home.
  - Relies on the **Marshall-Lerner condition** to appreciate the RER and **indirectly** boost real incomes.
  - Unwinds through capital outflows once the spike has passed: depreciate and export more; re-times the loss, does not mitigate it in PDV terms.
  - **Stabilized real income and consumption for constrained households**; stronger capital inflow and weaker export and RER co-movement with import prices: **stabilized terms of trade**.
- **FX policy as an anti-inflation policy**: more effective than monetary policy in protecting real incomes of constrained households and fighting imported inflation.
- Works by driving **interest parity wedges**. Monetary policy cannot. The interest parity condition limits any real income protection that monetary policy can deliver in isolation.

Results and Contributions

- **Normative analysis.** Jointly optimal monetary and FX policy response to a ToT shock.
  - **FX policy leads with real-income protection and disinflation. Indirect income effects** that rely on the Marshall-Lerner condition. Role increasing in the essentiality of imports, share of constrained.
  - **Monetary policy**: weaker role under final-good imports; employment-import complementarity: role increasing and complementary to FX policy. **More contractionary than RA due to a redistributive price effect**: import-induced depreciations redistribute from constrained wage earners to unconstrained net borrowers.
- **Quantitative two-agent model.** Nominal wage rigidity, financial intermediation frictions. Utilitarian-optimal FX interventions and monetary policy under perfect-foresight transitions.
- **Application: energy-price shocks.** Japan calibration, 2022-like price path:
  - **Joint optimum vs conventional:** energy 19.49% cheaper; 29.03% of real wage loss undone. Average inflation: 0.60% → 0.23%
  - **FX policy alone:** 26.64% of wage loss undone; average inflation: 0.60% → 0.28%
  - **Monetary policy alone:** 0.27% of wage loss undone; average inflation: 0.60% → 0.58%

Literature Connections

- **Conventional open-economy policy.** Free-floating exchange rate with undistorted capital flows; monetary policy targets nominal rigidities.
  Friedman (1953); Galí and Monacelli (2005); Bodenstein, Erceg and Guerrieri (2008)
- **ToT shocks in RA models and monetary policy.** Accommodation after energy and tariff shocks; energy importers: stable wage growth. Float optimal under dollar pricing.
  Lorenzoni and Werning (2023); Egorov and Mukhin (2023); Bianchi and Coulibaly (2025); Drechsel et al. (2026)
- **Terms-of-trade and energy shocks in TANK/HANK.** Monetary transmission and household incidence; capital flows left undistorted.
  Auclert et al. (2024*a*); Chan, Diz and Kanngiesser (2024); Acharya, Challe and Corsetti (2026)
- **HA real-income transmission in closed and open economies.** Monetary policy reaches constrained households through activity-induced indirect income effects. Real income channel of exchange rates.
  Kaplan, Moll and Violante (2018); Auclert (2019); Auclert et al. (2024*b*)
- **Intermediation frictions and FX interventions.** Study exchange rates, FX policy for closing intermediation frictions, stabilizing exchange rates under capital flow volatility and exchange rate objectives at the zero lower bound.
  Gabaix and Maggiori (2015); Amador et al. (2020); Fanelli and Straub (2021); Itskhoki and Mukhin (2023)

A Minimal Analytical Framework

- Two-period, two-agent framework. Real model, no nominal rigidity or monetary policy. **Goal: isolate the role of FX policy under heterogeneity.**
- Open economy: small in asset and import markets; large in export market.
- **Domestic output** Y_t: produced using labor and an imported input M_t. Output is tradeable; consumed at home or exported abroad:

$$C_t + X_t = Y_t$$

- **Foreign good (dollars/consumption basket)**: not produced or consumed at home; world numeraire.
- **Imported input**: price P_t^{T*} in foreign units.
- **Asset markets**: economy accesses international asset markets, assets **denominated in foreign good units**; world real interest rate R^*.
- Economy consumes domestic good only. P_t: nominal price and price level; e_t: nominal exchange rate. **Real exchange rate**: $\epsilon_t = e_t/P_t$, relative price of foreign good in domestic units.
- **Terms of trade**: $\epsilon_t P_t^{T*}$, relative price of the imported input in domestic units.

Firms and Production Technology

- Domestic output is produced with labor and the imported input using a Leontief technology:

$$Y_t = A \cdot \min\left\{N_t, \frac{M_t}{\theta}\right\}$$

- Cost function: $\mathcal{C}_t = w_t N_t + \epsilon_t P_t^{T*} M_t$
- Cost minimization implies $M_t = \theta N_t$, $Y_t = A N_t$. Marginal cost: $\mathcal{MC}_t = w_t + \epsilon_t P_t^{T*} \theta$
- Zero-profit condition: $Y_t = w_t N_t + \epsilon_t P_t^{T*} \theta N_t$ which gives:

$$w_t = A - \epsilon_t P_t^{T*} \theta$$

- A unit of labor with θ units of the imported input produces A units of output, of which w_t accrues to the worker and $\epsilon_t P_t^{T*} \theta$ paid to buy the imported input.
- A higher import price or a real depreciation lowers real wage one-for-one to restore zero profits in equilibrium.

Export Demand and the Marshall-Lerner Condition

- Downward-sloping demand for exports. World demand for the domestic good:

$$X_t = \zeta \epsilon_t^{1+\chi}$$

- $(1+\chi)$ is the price elasticity of export demand, with $\chi > 0$: relatively price-elastic demand.
- Export revenue in foreign units:

$$\frac{X_t}{\epsilon_t} = \zeta \epsilon_t^{\chi}$$

- **Marshall-Lerner condition**: $0 < \chi < \infty$, a real depreciation required to generate higher foreign purchasing power.

Households

- Two household types: **Hand-to-mouth**: mass $\lambda \in [0, 1)$. **Ricardian**: mass $1 - \lambda$.
- Consumption: C_t^p, C_t^r. Ricardians participate in asset markets. Asset position: f^*.
- Both types supply N units of labor inelastically.
- **FX policy**: capital control tax τ on Ricardian positions, rebated lump-sum T.
- No fiscal redistribution: HtMs receive no direct transfers. Isolates how FX policy reaches constrained households *indirectly*.

Ricardians (mass $1 - \lambda$)

$$\max_{C_t^r, f^*} \; u(C_1^r) + \beta\, u(C_2^r) \quad \text{s.t.}$$

$$C_1^r + (1+\tau)\,\epsilon_1\, f^* = w_1 N + T$$
$$C_2^r = w_2 N + \epsilon_2\, R^*\, f^*$$

Euler equation:

$$\underbrace{\frac{u'(C_1^r)}{\beta u'(C_2^r)}}_{\text{Domestic real rate}} = \underbrace{\frac{1}{(1+\tau)}}_{\text{Wedge}} \underbrace{R^* \frac{\epsilon_2}{\epsilon_1}}_{\text{Effective world real rate}}$$

Hand-to-Mouth (mass λ)

- No asset market access.
- Consume current incomes each period:

$$C_t^p = w_t N$$

Balance of Payments

- Period-by-period external budget constraints:

$$P_1^{T*}\theta N + F^* = \zeta\epsilon_1^X$$
$$P_2^{T*}\theta N = \zeta\epsilon_2^X + R^* F^* \qquad F^* = \frac{f^*}{1-\lambda}$$

- BoP identity: Trade balance = Net capital outflows $\equiv NX_t$:

$$\underbrace{NX_t}_{\text{Trade balance}} = \underbrace{\zeta\epsilon_t^X}_{\text{Export revenue}} \underbrace{- P_t^{T*}\theta N}_{\text{Import bill}}$$

- Aggregate intertemporal budget constraint:

$$\underbrace{P_1^{T*}\theta N + \frac{1}{R^*} P_2^{T*}\theta N}_{\text{PDV of import bills}} = \underbrace{\zeta\epsilon_1^X + \frac{1}{R^*}\zeta\epsilon_2^X}_{\text{PDV of export revenues}}$$

- Leontief + inelastic labor supply = perfectly inelastic import demand.

► Equilibrium definition

A Utilitarian Ramsey Planner's Problem

- Maximize population-weighted social welfare:

$$\max_{\{\epsilon_t\}} \lambda\big[u(C_1^p)+\beta u(C_2^p)\big]+(1-\lambda)\big[u(C_1^r)+\beta u(C_2^r)\big] \quad \text{s.t.}$$

$$NX_1(\epsilon_1)+\frac{1}{R^*}NX_2(\epsilon_2)=0$$

where

$$Y_t^N \equiv \big(A-\epsilon_t P_t^{T*}\theta\big)N \qquad \text{real labor income}$$
$$C_t^p = Y_t^N$$
$$C_t^r = Y_t^N - \frac{\epsilon_t NX_t}{1-\lambda}$$

- Primal formulation: Ricardian Euler equation not an implementability constraint.
- **Lemma.** The solution to this problem is implementable as a decentralized equilibrium with an appropriate FX policy (τ, T).

▸ Optimality conditions

Optimal FX Policy

- Assume $\beta R^* = 1$. Laissez-faire equilibrium: $\tau = 0$.

Proposition: Representative agent ($\lambda = 0$)

For **any** $(P_1^{T*}, P_2^{T*}) > 0$, the optimal allocation involves **no intertemporal distortions**: $\tau = 0$.

- Optimal policy is laissez-faire under RA: all agents smooth consumption perfectly given any price path.
- Special case: no ToT manipulation to internalize export market power.

Proposition: Heterogeneous agents ($\lambda \in (0, 1)$)

If $P_1^{T*} = P_2^{T*}$, then $\tau = 0$ for any λ.
If $P_1^{T*} > P_2^{T*}$, with $(\epsilon_t^{lf}, Y_t^{N,lf})$ laissez-faire and $(\epsilon_t^{O}, Y_t^{N,O}, \tau^O)$ the constrained efficient allocation:

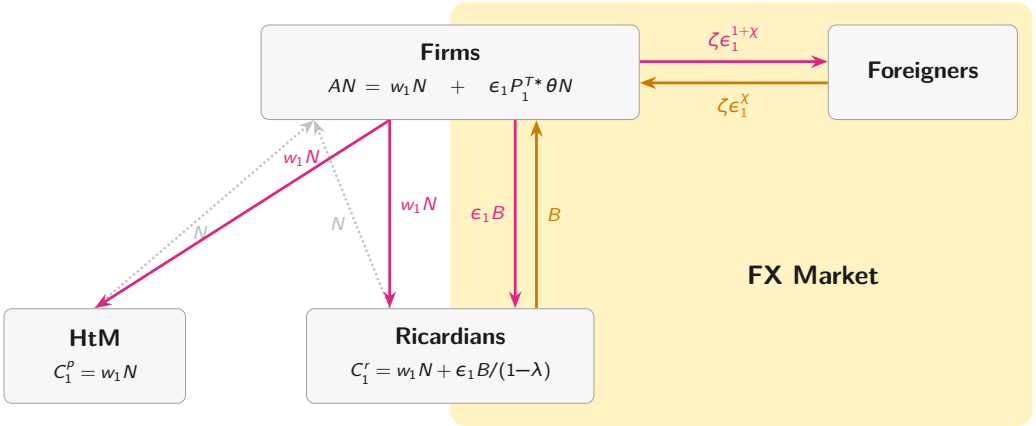
- **Weaker RER co-movement:** appreciation in response to the shock: $\epsilon_1^O < \epsilon_1^{lf}$, $\epsilon_2^O > \epsilon_2^{lf}$
- **Real income and ToT stabilization**: $\left|Y_2^{N,O} - Y_1^{N,O}\right| < \left|Y_2^{N,lf} - Y_1^{N,lf}\right|$, $|\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*}| < |\epsilon_1^{lf} P_1^{T*} - \epsilon_2^{lf} P_2^{T*}|$
- **More volatile capital flows**: $NX_1^O < NX_1^{lf} < 0 < NX_2^{lf} < NX_2^O$
- **Strictly positive intertemporal wedge**: $\tau^O > 0$

What Happened?

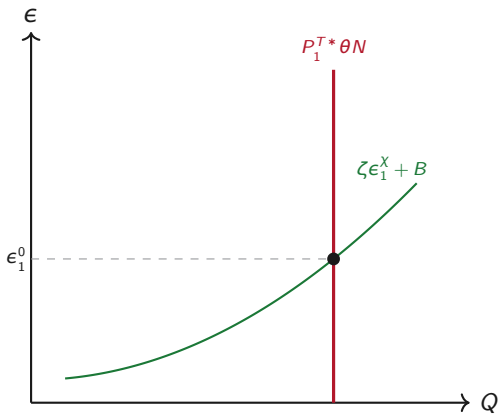
- Subsidize Ricardians to induce stronger net capital inflows in the high-price period. Import demand perfectly inelastic: capital inflows substitute for export revenues and appreciate the real exchange rate.
- Fewer exports and more resources retained at home at $t = 1$.
- At $t = 2$, after the spike has passed, must tolerate more depreciation and exports, larger capital outflows.
- **Why is this optimal? Smoother consumption path for HtM households through the shock.**
- No direct fiscal transfers: any additional borrowing induced by policy accrues to Ricardians and directly distorts their consumption path. **How does this policy deliver smoother consumption to HtM households then?**
- **The real income stabilization channel**: Capital-flow-induced RER appreciation raises real wages for **all** workers and **indirectly** boosts their purchasing power. Period 2 reverses this direction.
- **Marshall-Lerner condition** is the key: without this, policy cannot appreciate and boost real incomes.

Flow of Goods: The Foreign Exchange Market

- Suppose $P_1^T{}^* > P_2^T{}^*$ so that Ricardians are net borrowers at $t = 1$ with $B \equiv -F^*$.
- Ricardians and foreigners are net suppliers of the **foreign good** and demand the **domestic good**; firms are net demanders of the foreign good and supply the domestic good in exchange.



The FX Market

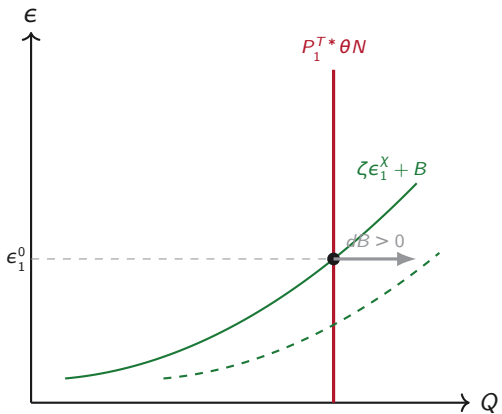


- BoP condition as foreign exchange market equilibrium: demand for and supply of the **foreign good**, with the real exchange rate as the market-clearing price:

$$\underbrace{P_1^T * \theta N}_{\text{Demand}} = \underbrace{\zeta\epsilon_1^X + B}_{\text{Supply}}$$

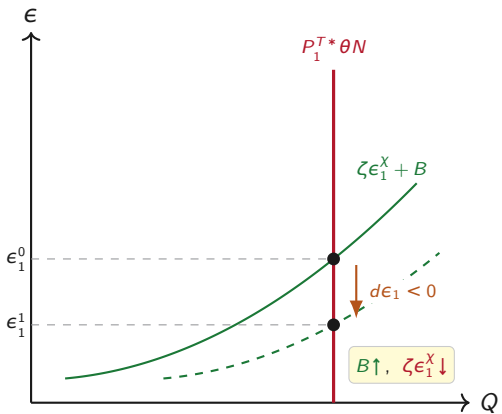
- Import demand perfectly inelastic.
- **Marshall-Lerner condition**: export revenue supply is upward-sloping: more foreign good earned through a real depreciation.

The FX Market: FX Policy Shifts Capital Inflows



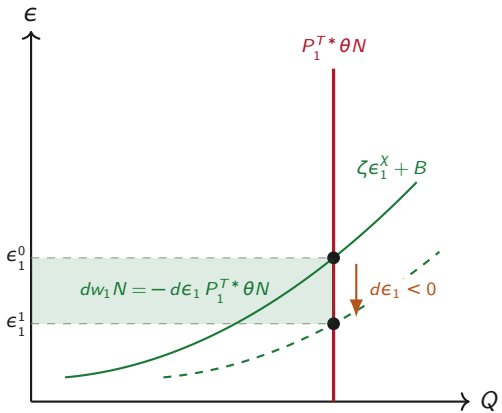
- **FX policy** induces stronger net capital inflows: Ricardians borrow more, $dB > 0$.
- The supply of foreign good in the FX market increases.
- More foreign purchasing power available to pay the import bill.

The FX Market: Real Appreciation



- Excess supply makes Ricardians bid down the real exchange rate.
- Foreigners **crowded out**: export revenues fall.
- The RER appreciates: $d\epsilon_1 < 0$.

The FX Market: Real Wage Gains



- A real appreciation lowers marginal costs for firms: positive profit opportunities arise that induce them to expand their import demand.
- Labor and imports are perfect complements: labor demand expands as firms compete for workers. With labor supply given, firms **bid the real wage up** until zero profits are restored: $w_1 = A - \epsilon_1 P_1^T * \theta$.
- **Every worker gains** through an indirect real income effect: capital inflows appreciate the RER and boost the real income of every worker, including the constrained.
- Marshall-Lerner elasticity crucial: income effect disappears as $\chi \to \infty$. Stronger as $\chi \to 0$.

Taking Stock: FX Policy and Real Income Stabilization

- Borrowing constraints and the Marshall-Lerner condition turn the exchange rate from a **shock absorber into a shock amplifier**: given an essential-import price spike, constrained households face amplified current real income losses due to a concurrent depreciation of the real exchange rate.
- **The real-income stabilization channel of exchange rate policy**: induce stronger capital inflows than laissez-faire to lean against a concurrent depreciation during the spike. Real wages rise for all workers: gains reach constrained households through indirect income effects.
- Policy re-times, does not eliminate the PDV loss: tolerates more depreciation, higher exports and lower real wages after the spike. Achieves a smoother consumption path for constrained households.
- Works by driving **interest parity wedges**. Optimal real rate endogenous.
- Does not require nominal rigidities and works independently of the level of activity: output and employment.

Adding Elastic Labor Supply and Monetary Policy

- Introduce elastic labor supply: N_t hours of employment. Both household types work the same amount of time, with disutility $v(N_t)$.
- Households do not choose hours. Nominal aggregate demand and monetary policy together determine aggregate labor supply.
- **Rigid nominal wage** $\overline{W}$. Phillips curve: $P_t = \overline{W}/w_t$.
- **Monetary policy** sets the money supply, $\mathcal{M}_t$, and the exchange equation determines nominal aggregate demand:

$$\mathcal{M}_t = P_t Y_t = P_t A N_t$$

- Monetary policy and the Phillips curve together determine the price level and employment. Real wage determined by firms' ZPC as before.
- With rigid nominal wages: a lower real wage is equivalent to a higher price level and vice versa.

The Ramsey Planner's Problem Revisited

- The utilitarian Ramsey planner now sets monetary and FX policy jointly:

$$\max_{\{N_t, \epsilon_t\}} \lambda\big[u(C_1^p) + \beta u(C_2^p)\big] + (1-\lambda)\big[u(C_1^r) + \beta u(C_2^r)\big] - \big[v(N_1) + \beta v(N_2)\big] \quad \text{s.t.}$$

$$NX_1(\epsilon_1, N_1) + \frac{1}{R^*} NX_2(\epsilon_2, N_2) = 0$$

where

$$Y_t^N = \big(A - \epsilon_t P_t^{T*} \theta\big) N_t, \qquad C_t^p = Y_t^N, \qquad C_t^r = Y_t^N - \frac{\epsilon_t NX_t}{1-\lambda}$$

- Primal formulation: Ricardian Euler equation and the nominal exchange equations are not implementability constraints.
- Nominal rigidity introduces non-neutrality: has no real costs by itself, but allows monetary policy to distort the **labor wedge** and influence the real economy through employment and output.
- **Lemma.** The solution to this problem is implementable as a decentralized equilibrium with an appropriate fiscal-monetary policy $(\tau, T, \mathcal{M}_t)$.

► Optimality conditions

Joint Optimal Policy: RA

Proposition: Representative agent ($\lambda = 0$)

For any $(P_1^{T*}, P_2^{T*}) > 0$, the optimal allocation involves no intertemporal distortions: $\tau = 0$. The labor wedge is distorted to account for the **static terms-of-trade correction**:

$$v'(N_t) = u'(C_t)\left(w_t - \frac{\epsilon_t P_t^{T*}\theta}{\chi}\right)$$

- Leontief: ToT externality corrected through labor wedge; planner internalizes static market power over exports: each additional unit of labor requires imported inputs, which depreciates the RER and worsens the terms of trade.
- Laissez-faire FX policy: all households smooth consumption optimally, no reason for policy to distort financial flows.

Joint Optimal Policy: HA

Proposition: Heterogeneous agents ($\lambda \in (0,1)$)

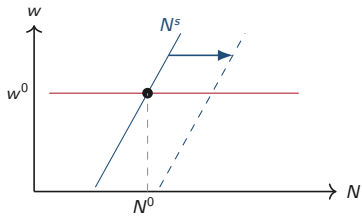
- $P_1^{T*} = P_2^{T*}$: no intertemporal distortions; labor wedge corrects for the terms-of-trade effect.
- $P_1^{T*} > P_2^{T*}$: positive intertemporal wedge, $\tau > 0$; labor wedge corrects for both the static terms-of-trade effect and a **redistributive price effect**:

$$v'(N_t) = \bar{u}_t' \left(w_t - \frac{\epsilon_t P_t^{T*} \theta}{\chi} \right) + \lambda \left[u'(C_t^p) - u'(C_t^r) \right] N X_t \left. \frac{d\epsilon_t}{dN_t} \right|_{NX_t}$$

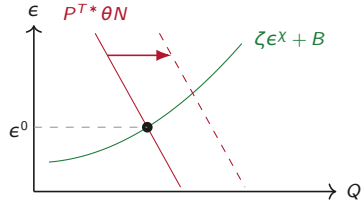
- When $P_1^{T*} > P_2^{T*}$, the redistributive price effect creates an additional motive to restrain employment and imports: optimal monetary policy more contractionary than an RA-equivalent rule.
- FX policy implements a positive intertemporal wedge as before: subsidizes Ricardians to induce stronger capital inflows during the high-price period.
- With FX policy active, monetary policy's role purely static: does not influence capital flows.

A Marginal Monetary Expansion: More Employment, More Imports

- Suppose $P_1^{T*} > P_2^{T*}$, Ricardians borrow: $B = -NX_1 > 0$. Consider a monetary expansion: $dN_1 > 0$.
- Labor supply expands: downward pressure on real wages lowers marginal costs and opens profit opportunities. Firms expand to hire the additional labor.
- **Import demand expands.** Labor and imports are perfect complements: firms also demand more imported inputs.
- Hold $B = -NX_1$ fixed: additional imports financed by current exports. General equilibrium with FX policy response endogenous: at the margin, indifferent between financing additional imports by current exports or capital inflows. Envelope theorem argument: employment's marginal impact purely static.



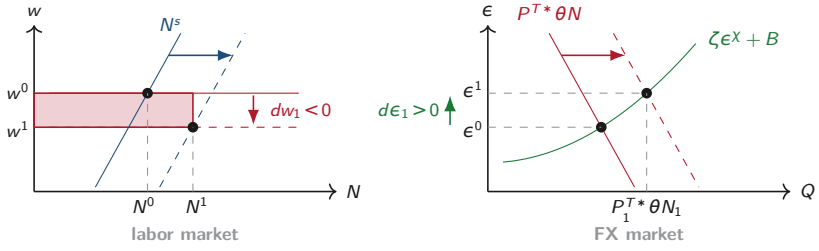
labor market



FX market

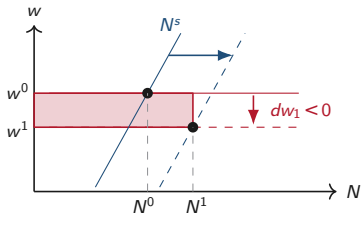
A Marginal Monetary Expansion: Depreciation and the Terms of Trade

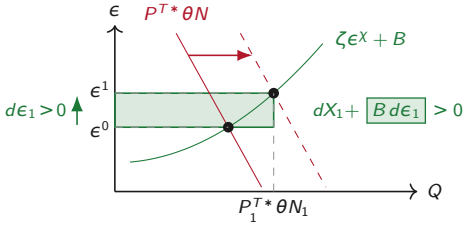
- Marshall-Lerner: foreigners only supply the requisite purchasing power at a lower world price. RER **depreciates**, $d\epsilon_1 > 0$ and exports rise, $dX_1 > 0$.
- Depreciation raises the marginal cost of imports. Zero profits: as the RER depreciates, real wage falls for every worker, $dw_1 < 0$.
- **Terms-of-trade externality**: employment expansion increases exports by depreciating the RER, and lowers the real wage. Firms do not internalize this, planner does; present even in an RA economy.



A Marginal Monetary Expansion: Redistributive Price Effect

- An import-induced depreciation also creates a **pecuniary** benefit for Ricardians: allows them to buy more of the domestic good, $B\,d\epsilon_1$, for the given amount of net foreign unit inflows, B.
- **Asymmetric gains**: real wages lost by workers finance this gain. In RA, this effect cancels out. In HA, **all borrowers are workers but not all workers are borrowers**.
- HtMs bear a lower wage: a net endogenous **transfer from HtMs to Ricardians**.
- Planner internalizes this asymmetric **redistributive price effect** of a monetary expansion and has incentive to restrain expansion beyond an RA-equivalent rule in the high import-price period.





Taking Stock: Joint Optimal Policy

- Relative to RA: during spike episodes, **FX policy induces stronger capital inflows** and **monetary policy is more contractionary**.
- FX policy distorts the **intertemporal wedge**. Monetary policy distorts the static **labor wedge** within each period and under employment-import complementarity, influences import demand.
- Both influence the FX market through different channels and thereby influence the equilibrium RER.
- FX policy seeks appreciation and higher real wages through capital inflows. Realized appreciation depends on how much monetary policy allows employment to expand.
- Feasible: FX policy induces capital inflows, monetary policy allows higher employment and imports, but must tolerate more depreciation and lower real wages; ToT externality hurts workers, redistributive price effects hurt constrained households.
- Also feasible: monetary policy restrains employment. FX-policy-induced capital inflows appreciate the RER and raise real wages: lower prices under rigid nominal wages.
- **Optimal**: real income gains to constrained through **higher wages, restrained employment**.

► Comparative statics: CES ► Comparative statics: final-good imports

Monetary Policy in Isolation

- Without FX policy, monetary policy faces the Ricardian Euler equation, i.e., the **interest parity condition** as an implementability constraint:

$$\frac{u'(C_1^r)}{\beta\, u'(C_2^r)} = R^* \frac{\epsilon_2}{\epsilon_1}$$

- Ricardian budget constraints and the Euler equation imply the Ricardian **saving function**:

$$F^* = F^*(\epsilon_1, \epsilon_2,\ N_1, N_2)$$

- Monetary policy sets the employment path, $\{N_1, N_2\}$; capital flows and the RER path follow as a consequence of balance of payments:

$$P_1^T * \theta N_1 + F^*(\epsilon_1, \epsilon_2,\ N_1, N_2) = \zeta \epsilon_1^X$$
$$P_2^T * \theta N_2 = \zeta \epsilon_2^X + R^* F^*(\epsilon_1, \epsilon_2,\ N_1, N_2)$$

- FX policy breaks this dependency: moves F^* independently of the employment path through interest parity wedges: tighter grip over the RER, independent of the level of real activity.

Quantitative Model

- Baseline: two-agent infinite-horizon framework.
- Segmented asset markets: Ricardians cannot access international assets directly. Domestic assets: nominal units; international assets: foreign-good units.
- Ricardians solve:

$$\max_{\{C_t^r, B_{t+1}^{M,r}\}} \sum_{t=1}^{\infty} \beta^{t-1}\left[u(C_t^r) - v(N_t)\right] \qquad \text{s.t.}$$

$$P_t C_t^r + B_{t+1}^{M,r} = W_t N_t + R_t^M B_t^{M,r} + T_t$$

- Optimality:

$$\frac{u'(C_t^r)}{\beta u'(C_{t+1}^r)} = R_{t+1}, \qquad \lim_{T\to\infty} \frac{B_{T+1}^r}{\prod_{s=2}^{T} R_s} = 0, \qquad B_{t+1}^r = \frac{B_{t+1}^{M,r}}{P_t}$$

- $R_{t+1} = \frac{R_{t+1}^M}{\Pi_{t+1}}$: real interest rate in domestic units; $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$ gross inflation rate.

Nominal Wage Rigidity and the Labor Wedge

- Adapted from Erceg, Henderson and Levin (2000) and Auclert, Rognlie and Straub (2024).
- Final labor service produced by an aggregator using $k \in [0, 1]$ tasks.
- For each task k, a union sets the nominal wage $W_t(k)$ for a continuum of workers.
- Each union faces a **private utility cost** of changing the nominal wage.
- Wage Phillips curve with symmetric unions:

$$v'(N_t) = (1 - \tau_t^N) w_t \bar{u}_t',$$

$$1 - \tau_t^N = \frac{1}{\mu_w} + \frac{\left(\frac{w_t}{w_{t-1}}\Pi_t - 1\right)\frac{w_t}{w_{t-1}}\Pi_t - \beta\left(\frac{w_{t+1}}{w_t}\Pi_{t+1} - 1\right)\frac{w_{t+1}}{w_t}\Pi_{t+1}}{\frac{\varepsilon_w}{\psi_{\mathrm{nr}}} w_t N_t \bar{u}_t'}$$

International Financial Intermediaries

- Adapted from Gabaix and Maggiori (2015).
- Take positions in domestic and international asset markets s.t. a balance sheet constraint: $\frac{F_{t+1}}{e_t} + F^*_{t+1} = 0$.
- Intermediary returns:

$$V_{t+1} = \left[R^* F^*_{t+1} + R^M_{t+1} \frac{F_{t+1}}{e_{t+1}} \right] = \left[R^* - R^M_{t+1} \left(\frac{e_t}{e_{t+1}} \right) \right] F^*_{t+1}$$

- Seek to maximize returns subject to a limited commitment friction:

$$\max \frac{1}{R^*} V_{t+1} \quad \text{s.t.} \frac{1}{R^*} V_{t+1} \geq \Gamma F^{*2}_{t+1}$$

- Optimal asset position a function of the interest parity gap:

$$F^*_{t+1} = \frac{1}{\Gamma} \left[1 - \frac{R^M_{t+1}}{R^*} \left(\frac{e_t}{e_{t+1}} \right) \right] = \frac{1}{\Gamma} \left[1 - \left(\frac{R_{t+1}}{R^*} \frac{\epsilon_t}{\epsilon_{t+1}} \right) \right]$$

Fiscal-Monetary Policy and Accounting

- Monetary policy sets the money growth rate. Nominal aggregate demand given by the exchange equation:

$$g_t^{\mathcal{M}} = \Pi_t \frac{Y_t}{Y_{t-1}}$$

  Equivalently, monetary policy determines the ex post real rate R_1 and forward-looking nominal interest rates:

$$R_{t+1}^M = R_{t+1}\Pi_{t+1} \quad \text{for } t \geq 1$$

- Fiscal policy: domestic and foreign asset positions, with budget constraint:

$$B_{t+1}^{M,G} + e_t B_{t+1}^{G*} + (1-\lambda)T_t = R_t^M B_t^{M,G} + e_t R^* B_t^{G*}$$

$$\lim_{T\to\infty} \frac{B_{T+1}^G}{\prod_{s=2}^T R_s} = 0, \qquad \lim_{T\to\infty} \frac{B_{T+1}^{G*}}{R^{*T-1}} = 0, \qquad B_{t+1}^G \equiv \frac{B_{t+1}^{M,G}}{P_t}$$

- Bond-market clearing:

$$B_{t+1} + B_{t+1}^G = \epsilon_t F_{t+1}^*$$

- Balance of payments:

$$P_t^{T*} M_t + F_{t+1}^* + B_{t+1}^{G*} = \zeta \epsilon_t^X + R^*(F_t^* + B_t^{G*}) - R^* \Gamma F_t^{*2}$$

Ramsey Problem

$$\max_{\{N_t,\epsilon_t,\tilde{R}_{t+1},C_t^r,C_t^p\}_{t=1}^{\infty}} \sum_{t=1}^{\infty}\beta^{t-1}\left[\lambda u\left(C_t^p\right)+(1-\lambda)u\left(C_t^r\right)-v(N_t)\right] \qquad \text{s.t.}$$

$$\underbrace{\sum_{t=1}^{\infty}\frac{\frac{1}{\Gamma}\left(1-\frac{\tilde{R}_{t+1}}{R^*}\right)^2}{R^{*\,t-1}}}_{\text{carry trade cost}}+\sum_{t=1}^{\infty}\frac{P_t^{T*}\theta N_t}{R^{*\,t-1}}=\sum_{t=1}^{\infty}\frac{\zeta\epsilon_t^{\chi}}{R^{*\,t-1}}+R^*B_1^{G*}+R_1\frac{\epsilon_0F_1^*}{\epsilon_1}$$

$$\tilde{R}_{t+1}=\frac{u'\left(C_t^r\right)\epsilon_t}{\beta u'\left(C_{t+1}^r\right)\epsilon_{t+1}}, \qquad C_t^r=A_tN_t-\frac{\zeta\epsilon_t^{1+\chi}}{1-\lambda}+\frac{\lambda}{1-\lambda}\epsilon_tP_t^{T*}\theta N_t, \qquad C_t^p=(A_t-\epsilon_tP_t^{T*}\theta)N_t$$

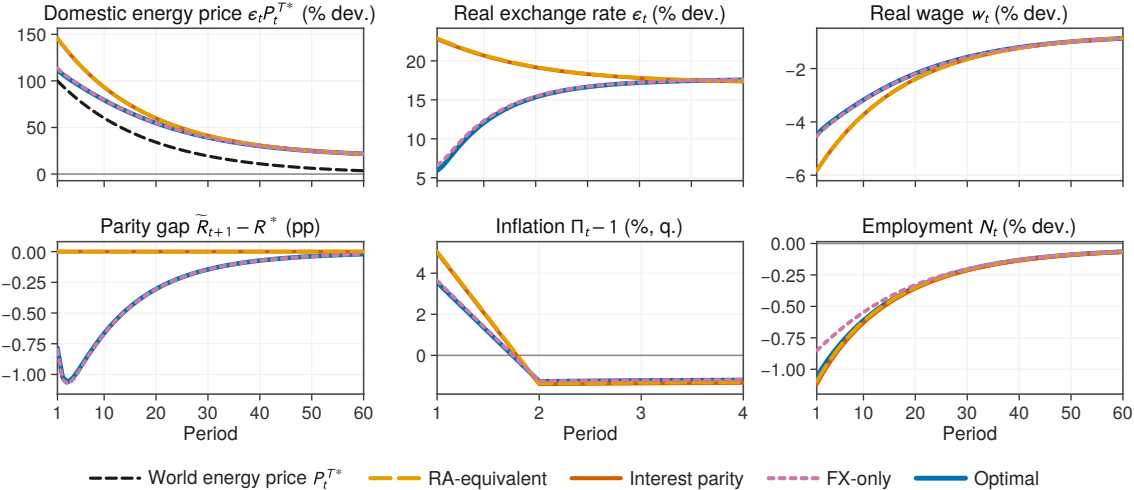
- Qualitatively the same primal problem as the analytical framework, except the carry cost term in IBC: interest parity wedges are costly in the form of carry trade costs from intermediary profits.
- Analytical framework nested: $\Gamma \to 0$, the same primal problem, decentralized with capital controls.
- **Lemma.** The solution to this problem is implementable as a decentralized equilibrium.
- Phillips curve not an implementability constraint: costless non-neutrality; inflation path implements the optimal labor wedge to satisfy the Phillips curve.

Application: Energy Price Shocks

- Essential import: fossil fuels. Calibrated to Japan: 98% import dependence; over 4% of output spent on fossil fuels in 2022.
- Compare jointly optimal policy under perfect foresight to three counterfactual policies:
  - **RA-equivalent**: no intertemporal distortions; labor wedge corrects terms-of-trade distortion.
  - **Interest parity**: FX policy closes the financial intermediation friction; implements optimal monetary policy in isolation.
  - **FX-only**: intertemporal stance free; labor wedge corrects RA-like terms-of-trade distortion only.
- Policy response under two import price paths:
  - **MIT shock**: perfect-foresight impulse response.
  - **2022 spike**: response to a deterministic 2022-like energy-price path.

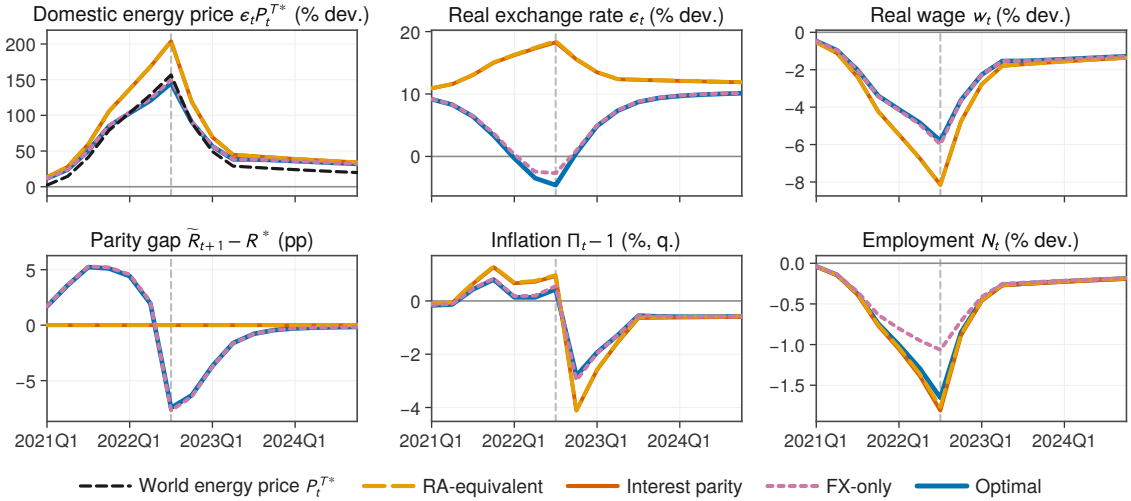
► Calibration

Transitory Spike



► Additional outcomes ► Output shares

2022 Spike



► Additional outcomes ► Output shares

2022 Spike: Policy Comparison

2021–23 path	RA-equiv.	Interest parity	FX-only	Optimal
Peak energy-price amplification factor	1.30	1.30	0.96	0.92
Peak real-wage loss undone (%)	–	0.27	26.64	29.03
Peak domestic energy price (% lower)	–	0.18	17.88	19.49
Avg. quarterly inflation through peak (%)	0.60	0.58	0.28	0.23

- At the peak of the crisis, heterogeneity accounts for 71.06% of the contemporaneous depreciation under RA-equivalent policy relative to a world with no HtM households.
- Monetary policy set optimally, with the intertemporal stance suppressed, does little on its own in protecting HtM households: faces the interest parity condition.
- **FX policy leads with real income protection and disinflation. Monetary policy supports with a disinflationary stance**.
- Optimal policy re-times when the country pays for real income losses, does not eliminate them: present value of aggregate consumption differs by only 0.0005% between optimal and RA-equivalent policies.

Conclusion

- Borrowing constraints and the Marshall-Lerner condition turn the RER from a **shock absorber into a shock amplifier**: amplifies current real-income losses for constrained households during episodes of essential-import price spikes.
- A laissez-faire approach to exchange rate management is costly for constrained households. Optimal policy should **lean against a contemporaneous real depreciation** during these episodes.
- **The real-income stabilization channel**: capital inflows appreciate the RER and raise real wages for all workers. Gains reach constrained households through **indirect income effects**.
- With FX policy active, the role for monetary policy changes and becomes supportive. As opposed to a typical HANK prescription, optimal monetary policy is more contractionary than an RA-equivalent rule: a redistributive price effect, attributable to the Marshall-Lerner condition, is at play.
- **FX policy becomes an anti-inflation policy.** In a monetary economy, real income protection arrives in the form of lower inflation. FX policy is more effective than monetary policy in addressing import-price-spike-induced inflation.
- Monetary policy in isolation is limited by the interest parity restriction: it cannot drive intertemporal wedges in a small open economy. Most real income protection power of policy lives here.

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Appendix

Equilibrium Definition

Given an exogenous sequence $\{P_t^{T*}\}$ and exchange rate policy (τ, T), an equilibrium consists of $\{C_t^p, C_t^r, F^*, w_t, \epsilon_t\}$ such that:

- HtM households consume current income: $C_t^p = w_t N$
- Ricardian households optimize: $u'(C_1^r)\epsilon_1(1+\tau) = \beta R^* u'(C_2^r)\epsilon_2$
- Firms optimize: $w_t = A - \epsilon_t P_t^{T*}\theta$
- Fiscal budget holds: $T = \tau\epsilon_1 \frac{F^*}{1-\lambda}$
- Domestic goods market clears:

$$\lambda C_t^p + (1-\lambda)C_t^r + \zeta\epsilon_t^{1+\chi} = AN$$

- Balance of payments:

$$P_1^{T*}\theta N + F^* = \zeta\epsilon_1^{\chi}, \qquad P_2^{T*}\theta N = \zeta\epsilon_2^{\chi} + R^* F^*$$

Optimality Conditions

Optimal intertemporal wedge:

$$\epsilon_1 u'(C_1^r) + \Omega_1 = \beta R^* \left[\epsilon_2 u'(C_2^r) + \Omega_2\right]$$

$$\Omega_t \equiv \frac{\lambda\left[u'(C_t^p) - u'(C_t^r)\right]\left(\frac{\epsilon_t}{\chi} - NX_t \frac{d\epsilon_t}{dNX_t}\right)}{\frac{1+\chi}{\chi}}$$

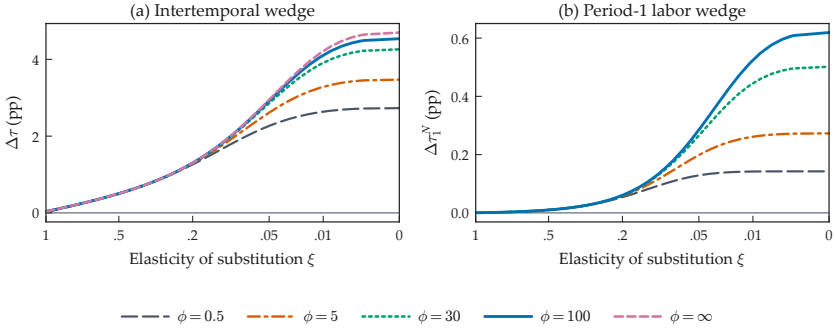
$$\tau = \frac{\Omega_1 - \beta R^* \Omega_2}{\epsilon_1 u'(C_1^r)}$$

Optimal labor wedge:

$$v'(N_t) = \bar{u}_t' \left(w_t - \frac{\epsilon_t P_t^{T*} \theta}{\chi}\right) + \lambda\left[u'(C_t^p) - u'(C_t^r)\right] NX_t \frac{d\epsilon_t}{dN_t}\Big|_{NX_t}$$

$$\tau_t^N = \frac{1}{\chi} \frac{\epsilon_t P_t^{T*} \theta}{w_t} - \frac{\lambda\left[u'(C_t^p) - u'(C_t^r)\right] NX_t \frac{d\epsilon_t}{dN_t}\Big|_{NX_t}}{\bar{u}_t' w_t}$$

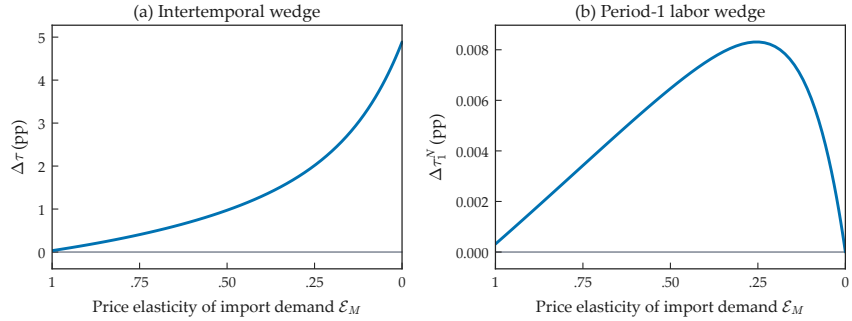
Comparative Statics: CES Production



- Generalize production to CES: $Y_t = A_t \left[(1-\alpha) N_t^{\frac{\xi-1}{\xi}} + \alpha \left(\frac{M_t}{\theta} \right)^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}}$.
- Comparative statics with respect to the price elasticity of import demand. Plotted case: $P_1^{T*} > P_2^{T*}$, optimal wedge differences between $\lambda > 0$ and $\lambda = 0$ economies. Increasing as $\xi \to 0$ and $\phi \to \infty$: **essential imports strengthen both policy motives** under heterogeneity.

◂ back

Comparative Statics: Imports as Final Goods



- Imported good as a final good with Stone-Geary preferences: $u(C, M) = \frac{\left[C^{1-\omega}(M-\underline{M})^{\omega}\right]^{1-\sigma}}{1-\sigma}$, $M \geq \underline{M}$.
- Comparative statics with respect to the price elasticity of import demand. Plotted case: $P_1^{T*} > P_2^{T*}$. Optimal wedge differences between $\lambda > 0$ and $\lambda = 0$ economies.
- **FX policy's role increasing with essentiality**: $\Delta\tau$ increases as import demand becomes inelastic.
- $\Delta\tau_1^N$ small, weaker role for monetary policy. Perfectly inelastic demand: heterogeneity adds no role.

◂ back

Labor Aggregator

- The competitive labor packer combines differentiated tasks into final labor:

$$N_t = \left(\int_0^1 N_{k,t}^{\frac{\varepsilon_w - 1}{\varepsilon_w}} \, dk \right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}}, \qquad \varepsilon_w > 1$$

- It solves

$$\min_{\{N_{k,t}\}_{k \in [0,1]}} \int_0^1 W_{k,t} N_{k,t} \, dk \quad \text{s.t.} \quad \left(\int_0^1 N_{k,t}^{\frac{\varepsilon_w - 1}{\varepsilon_w}} \, dk \right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}} \geq N_t$$

- Cost minimization yields

$$N_{k,t} = \left(\frac{W_{k,t}}{W_t} \right)^{-\varepsilon_w} N_t, \qquad W_t = \left(\int_0^1 W_{k,t}^{1-\varepsilon_w} \, dk \right)^{\frac{1}{1-\varepsilon_w}}$$

◂ back

Unions' Wage-Setting Problem and the WPC

- Taking the task-demand schedule as given, union k chooses its nominal wage sequence to solve

$$\max_{\{W_{k,t+s}\}_{s\geq 0}} \sum_{s\geq 0} \beta^s \left[\lambda u(C^p_{t+s}) + (1-\lambda) u(C^r_{t+s}) - v(N_{t+s}) - \frac{\psi_{\text{nr}}}{2} \left(\frac{W_{k,t+s}}{W_{k,t+s-1}} - 1 \right)^2 \right]$$

  subject to

$$N_{k,t+s} = \left(\frac{W_{k,t+s}}{W_{t+s}} \right)^{-\varepsilon_w} N_{t+s}.$$

- Each union is infinitesimal and takes W_{t+s} and N_{t+s} as given.
- In a symmetric equilibrium, $W_{k,t} = W_t$, $N_{k,t} = N_t$ for every k. Wage Phillips curve:

$$v'(N_t) = (1-\tau^N_t) w_t \bar{u}'_t,$$

$$1-\tau^N_t = \frac{1}{\mu_w} + \frac{\left(\frac{w_t}{w_{t-1}} \Pi_t - 1 \right) \frac{w_t}{w_{t-1}} \Pi_t - \beta \left(\frac{w_{t+1}}{w_t} \Pi_{t+1} - 1 \right) \frac{w_{t+1}}{w_t} \Pi_{t+1}}{\frac{\varepsilon_w}{\psi_{\text{nr}}} w_t N_t \bar{u}'_t}$$

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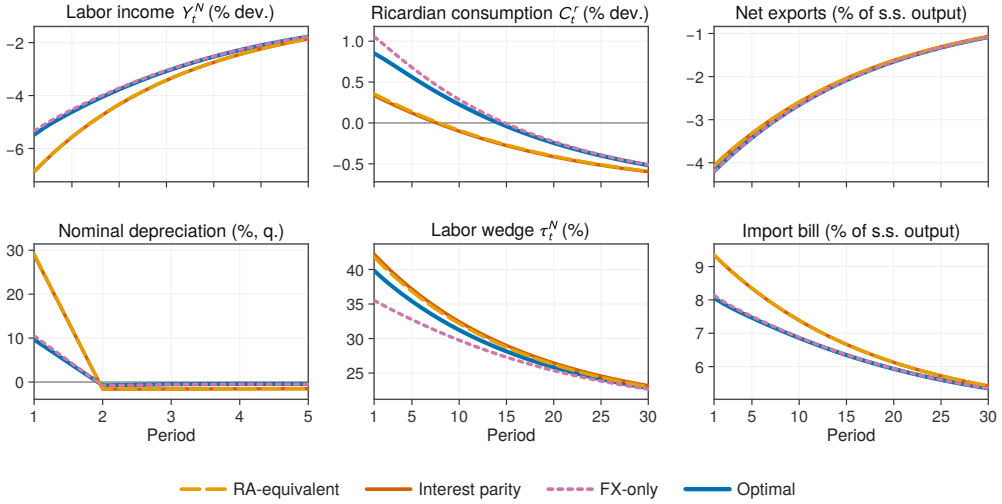
Calibration

Quarterly calibration for Japan

	Parameter	Value	Target or source
R^*	Gross world interest rate	1.0021	Net exports of −0.5% of GDP
β	Discount factor	0.9979	$\beta R^* = 1$
σ	Inverse intertemporal elasticity	4.0	Intertemporal elasticity of 0.25
η	Labor-disutility scale	0.9261	Steady-state employment normalization
ϕ	Inverse Frisch curvature	29.36	Output falls 1% on impact after a 100% import-price rise in the RA economy
λ	HtM household share	0.35	Kaplan, Violante and Weidner (2014) ballpark for US
ζ	Export demand scale	1.9758	Fossil-fuel import bill of 4%
χ	Marshall-Lerner elasticity	0.25	Modest short-run response: Rose (1991); Bahmani-Oskooee and Ratha (2004)
ρ_{pT*}	Import-price persistence	0.945	AR(1) on detrended IMF energy price index (IMF 2026)
Γ	Intermediation friction	9.0	Fanelli and Straub (2021)
ε_w	Wage substitution elasticity	6.129	Core CPI inflation of 0.097% per quarter (2014Q4–2019Q4) in RA steady state
ψ_{nr}	Nominal wage rigidity	10598.1	Core CPI inflation of 0.411% per quarter (2020Q4–2022Q3) in RA 2022 spike
B^{G*}_{ss}	Public foreign assets	60.0	Net foreign assets of 60% of annual GDP

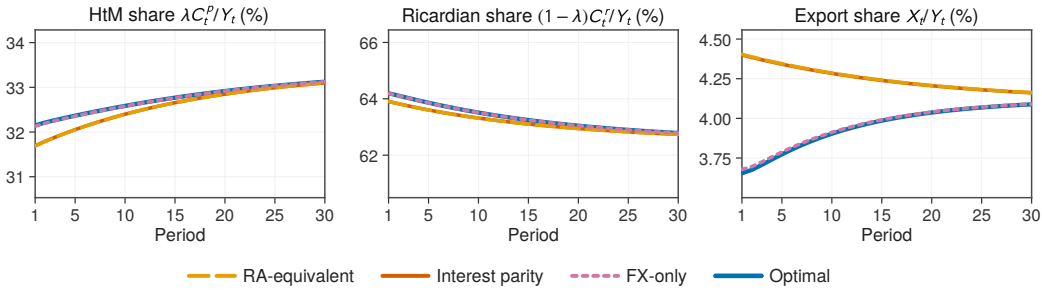
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Transitory Spike: Additional Outcomes



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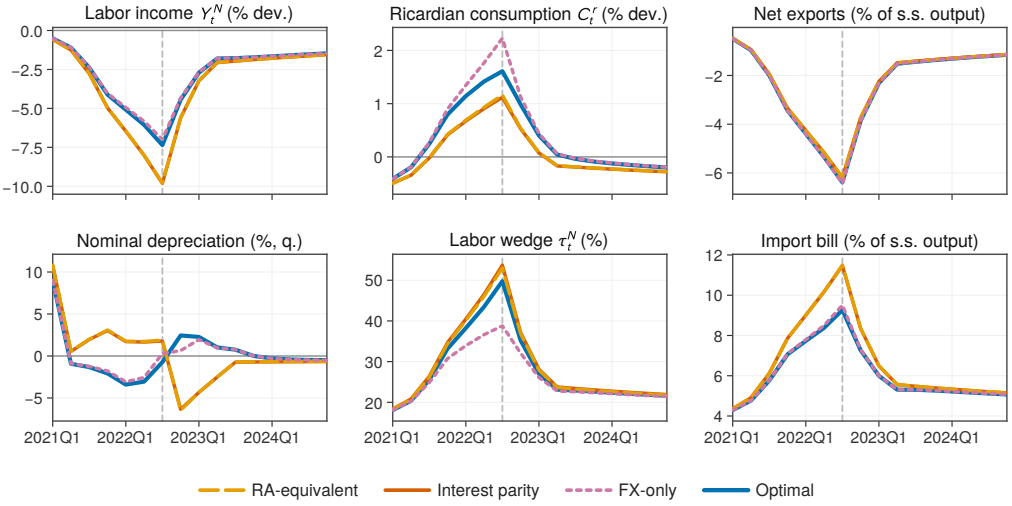
Transitory Spike: Output Shares



- During the spike, optimal policy retains more output for domestic consumption and reduces the export share. HtM share goes up due to indirect income effects.

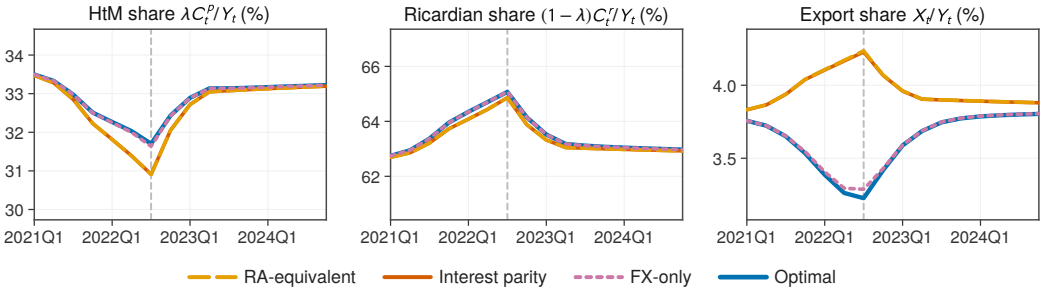
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2022 Spike: Additional Outcomes



◂ back

2022 Spike: Output Shares



- Around the peak of the crisis, optimal policy lowers the export share and retains more output for domestic consumption. HtM share goes up due to indirect income effects.